Privacy-preserving screening · implemented flow

Los Antimarrulla · IBM Z Datathon 2026 · src/screening/ · verified locally; LinuxONE s390x run pending

BEFORE THE DEMO (OFFLINE) · make model · make verify

CDC UCI 891
253,680 rows · 19 of 21 features used

→ 70/15/15 splits
stratified · seed 2026

→ Logistic regression
threshold selected on validation

→ Exported model (cdc19-v1 schema)
$W = \text{round}(10^9 \cdot w)$ · $B = \text{round}(10^9 \cdot b)$ · exponent 0 · hashes

PLANNED SAME-VM DEMO · trusted host administrator · SYNTHETIC demonstration patients only

Hospital container · owns the private key

1 · Split-screen UI
synthetic patient (19 features)

→ 2 · Validate schema
reject values outside integer ranges

↓ 3 · Encrypt with Paillier
2048-bit private key in memory

→ **TLS 1.3**
mutual (mTLS)

→ 4 · Validate message
versions, ≥2048-bit key, 19 group ciphertexts, overflow bound

↓ 5 · $Enc(z) = \prod Enc(x_i)^{W_i} \cdot Enc(B) \pmod{n^2}$
public weights only; rerandomize result

← **TLS 1.3**
mutual (mTLS)

← 6 · Decrypt integer z
own pending responses, once only

← 7 · Risk = $\sigma(z / 10^9)$
threshold + model contributions

↓ 8 · AES-256-GCM vault (unique nonce; ID as AAD)
32-byte key outside repository · result returned to UI

Risk service container · no private key

Logs: IDs, milliseconds and byte counts
no input values or decrypted results

Receives public key + 19 ciphertexts (≈24 KB)
Returns one encrypted score (≈1.4 KB)

Tests: decrypted integer equals the reference · floating-point logit error ≤ 9.55×10^{-8} · negatives, zeros, extremes, invalid inputs · export/reload · no private key at the service · no patient data in operational service logs.

Screening support, not diagnosis. ENSANUT and E5 are separate exploratory extras; Granite and CKKS are not implemented.